

Olivia S. Kowalczyk, PhD

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Neuroimaging scientist advancing methods to study the **brain and spinal cord**
and deriving **clinically translatable biomarkers** of central nervous system conditions.

EDUCATION

Doctor of Philosophy (PhD) in Neuroimaging	2017 – 2021
Institute of Psychiatry, Psychology & Neuroscience (IoPPN), King's College London	
Thesis: <i>Effects of ADHD medications on brain activation and functional connectivity in youth with ADHD during response inhibition and working memory.</i>	
BSc Psychology, First Class Honours	2013 – 2016
Royal Holloway, University of London	

PROFESSIONAL APPOINTMENTS

King's Prize Research Fellow	2023 – PRESENT
Department of Neuroimaging, IoPPN, King's College London	
SPM Training and Documentation Lead	2022 – PRESENT
Functional Imaging Laboratory (FIL), Department of Imaging Neuroscience, University College London	
Postdoctoral Research Associate	2021 – 2023
Department of Neuroimaging & Wolfson Centre for Age-related Diseases, IoPPN, King's College London	
Graduate Teaching Assistant	2018 – 2021
IoPPN, King's College London	
Research Assistant	2016 – 2017
Department of Neuroimaging, IoPPN, King's College London	
MRI Research Assistant	2015 – 2016
Imanova Ltd. (currently known as Perceptive)	

AWARDS & FUNDING

King's Prize Fellowship (£170,500)	2023
King's College London	
Open Research Award	2023
Institute of Psychiatry, Psychology & Neuroscience, King's College London	
Novel Neuroimaging Approaches and Applications (£30,000, co-investigator)	2023
NIHR Maudsley Biomedical Research Centre	
OHBM Open Research Award (as part of <i>The SPM Team</i>)	2023
Organization for Human Brain Mapping	
Summer Meeting Poster Prize	2021
British Association for Psychopharmacology	
SIPS Commendation (as part of <i>The RIOT Science Club</i>)	2021
Society for the Improvement of Psychological Science	
Health Sciences Doctoral Training Centre Research Theme Activities (£1,280)	2019
King's College London	
Summer Meeting Poster Prize	2018
British Association for Psychopharmacology	
4-year PhD Studentship (£101,000)	2017
NIHR Maudsley Biomedical Research Centre	
Passport Award for Extracurricular Activities and Community Action	2016
Royal Holloway, University of London	

INVITED TALKS & SCIENCE COMMUNICATION

SPM Homecoming and SPM25: New Developments in Neuroimaging Analysis Andy's Brain Tube (Dr Andrew Jahn's science communication platform)	2025
Open and Reproducible Science: A How-to Experimental and Translational Medicine Course, NIHR Maudsley Biomedical Research Centre	2025
Advances in Spinal Cord Imaging: Implications for Pain Research Pain Research Clinical Academic Workshop, King's Health Partners	2024
Experimental Design for fMRI SPM MRI Course, FIL, University College London	2023
Preparing for a PhD Viva Forensic & Neurodevelopmental Sciences PhD Forum, King's College London	2023
What Is Open Science and How Can I Use It in Addictions Research? Society for the Study of Addiction Early Career Research Network	2023
What Senior Academics Can Do to Support Reproducible and Open Research ReproducibiliTea, McMaster University Open Research: A Vision for the Future	2021

ORGANISED SYMPOSIA & WORKSHOPS

SPM MRI Course FIL, Department of Imaging Neuroscience, University College London	2022 - PRESENT (BI-ANNUAL)
30 Years of SPM: Celebrating the Evolution of Neuroimaging Analysis Organization for Human Brain Mapping	2025
SPM Homecoming FIL, Department of Imaging Neuroscience, University College London	2025
Introduction to Resting-State fMRI Analysis The RIOT Science Club	2021
Open Research: A Vision for the Future The RIOT Science Club	2021
fMRI Analysis for Beginners The RIOT Science Club	2020
King's Open Research Conference The RIOT Science Club & King's Open Research Group Initiative	2020

TEACHING EXPERIENCE

Themes: neuroimaging methods, research methods, open scholarship
Format: lectures, small group tutorials, workshops (online and in-person)
Level of study: undergraduate, postgraduate

SUPERVISION

Leda Bianchi, PhD Forensic & Neurodevelopmental Sciences	2024 – PRESENT
Mona Beidas, PhD Forensic & Neurodevelopmental Sciences	2024 – PRESENT
Joseph L. Taylor, PhD Neuroimaging	2023 – PRESENT
Alexandra Voce, BSc Neuroscience & Psychology (First Class Honours) 2024 British Association for Neuroscience Undergraduate Prize recipient	2023 – 2024
Chan Young Yang, MSc Organisational Psychiatry & Psychology (Distinction)	2022 – 2023
Alexander Eaton Turner, BSc Psychology (Upper Second Class Honours)	2019 – 2020

 SELECTED PUBLICATIONS

Spinal cord imaging

- Kowalczyk, O.S.**, Howard, M.A., & Lythgoe, D.J. (2025). Susceptibility-matched padding improves the quality of cervical and lumbar spinal fMRI. *bioRxiv*, 2025-12.
- Kowalczyk, O.S.**, Medina, S., Venezia, A., Tsivaka, D., Ahmed, A.I., Williams, S.C.R., Brooks, J.C.W., Lythgoe, D. J., & Howard, M.A. (2025). Test-retest reliability of sensorimotor activity measured with spinal cord fMRI. *bioRxiv*, 2025-09.
- Banerjee, R., Kaptan, M., Tinnermann, A., Khatibi, A., Dabbagh, A., Büchel, C., ..., **Kowalczyk, O.S.**, ..., & Cohen-Adad, J. (2025). EPISeg: Automated segmentation of the spinal cord on echo planar images using open-access multi-center data. *Imaging Neuroscience*, 3, IMAG-a.98.
- Valošek, J., Mathieu, T., Schlienger, R., **Kowalczyk, O.S.**, & Cohen-Adad, J. (2024). Automatic segmentation of the spinal cord nerve rootlets. *Imaging Neuroscience*, 2, 1–14.
- Kowalczyk, O.S.**, Medina, S., Tsivaka, D., McMahon, S.B., Williams, S.C.R., Brooks, J.C.W., Lythgoe, D.J., & Howard, M.A. (2024). Spinal fMRI demonstrates segmental organisation of functionally connected networks in the cervical spinal cord: A test–retest reliability study. *Human Brain Mapping*, 45(2), e26600.
- Tsivaka, D., Williams, S.C.R., Medina, S., **Kowalczyk, O.S.**, Brooks, J.C.W., Howard, M.A., Lythgoe, D.J., & Tsougos, I. (2023). A second-order and slice-specific linear shimming technique to improve spinal cord fMRI. *Magnetic Resonance Imaging*, 102, 151-163.

Neuroimaging methods development

- Zeidman, P., Ashburner, J., Barnes, G., **Kowalczyk, O.S.**, Lambert, C., Litvak, V., Nichols, T.E., Parr, T., Tierney, T.M., & Friston, K. (2025). SPM—30 years and beyond. *Cerebral Cortex*, 35(8), bhaf234.
- Tierney, T.M., Alexander, N.A., Ashburner, J., Avila, N.L., Balbastre, Y., Barnes, G., Bezsudnova, Y., Brudfors, M., Eckstein, K., Flandin, G., Friston, K., Jafarian, A., **Kowalczyk, O.S.**, Litvak, V., Medrano, J., Mellor, S., O’Neil, G., Parr, T., Razi, A., Timms, R., & Zeidman, P. (2025). SPM 25: Open source neuroimaging analysis software. *Journal of Open Source Software*, 10(110), 8103.
- Demetriou, L., **Kowalczyk, O.S.**, Tyson, G., Bello, T., Newbould, R.D., & Wall, M.B. (2018). A comprehensive evaluation of increasing temporal resolution with multiband-accelerated protocols and effects on statistical outcome measures in fMRI. *NeuroImage*, 176, 404-416.

Translational neuroimaging

- Taylor, J.L., Lawn, T., **Kowalczyk, O.S.**, Graven-Nielsen, T., Howard, M.A., & Bannister, K. (2025). Cuff algometry induces large yet variable conditioned pain modulation effects. *medRxiv*, 2025-09.
- El Masri, S., **Kowalczyk, O.S.**, Chiu, T.H., Criaud, M., Lukito, S., Mazibuko, N., Salazar de Pablo G., Perez Rodriguez V., Makos O., Lam S.L., Westwood S.J., Eaton-Turner A., Bozhilova N., Conti A., Santosh P., Roessner V., Kohls G., Mehta M.A., & Rubia, K. (2025). Differential effects of a single dose of lisdexamfetamine and guanfacine on cognitive function in children with ADHD. *Frontiers in Psychiatry*, 16, 1676472.
- Lukito, S., Lam, S.L., Criaud, M., Westwood, S., **Kowalczyk, O.S.**, Curran, S., Barrett, N., Abbott, C., Liang, H., Simonoff, E., Barker, G.J., Giampietro, V., & Rubia, K. (2024). Effects of fMRI neurofeedback of right inferior frontal cortex on inhibitory brain activation in children with ADHD. *Philosophical Transactions B*, 379(1915), 20230097.
- Howard, M.A., Lawn, T., & **Kowalczyk, O.S.** (2023). Harnessing the power of endogenous pain control mechanisms for novel therapeutics: How might innovations in neuroimaging help? *Current Opinion in Supportive & Palliative Care*, 17(3), 150-155.
- Kowalczyk, O.S.**, Cubillo, A.I., Criaud, M., Giampietro, V., O’Daly, O.G., Mehta, M.A., & Rubia, K. (2023). Single-dose effects of methylphenidate and atomoxetine on functional connectivity during an n-back task in boys with ADHD. *Psychopharmacology*, 240(10), 2045-2060.
- Westwood, S.J., Criaud, M., Lam, S.L., Lukito, S., Wallace-Hanlon, S., **Kowalczyk, O.S.**, ..., & Rubia, K. (2021). Transcranial direct current stimulation combined with cognitive training in adolescent boys with ADHD: A double-blind, randomised, sham-controlled trial. *Psychological Medicine*, 53(2), 497-512.
- Lam, S.L., Criaud, M., Lukito, S., Westwood, S.J., Agbedjro, D., **Kowalczyk, O.S.**, ... & Rubia, K. (2022). Double-blind, sham-controlled randomized trial testing the efficacy of fMRI neurofeedback on clinical and cognitive measures in children with ADHD. *The American Journal of Psychiatry*, 179(12), 947-958.
- Westwood, S.J., Bozhilova, N., Criaud, M., Lam, S.L., Lukito, S., Wallace-Hanlon, S., **Kowalczyk, O.S.**, Kostara, A., Mathew, J., Wexler, B.E., Cohen Kadosh, R., Asherson, P., & Rubia, K. (2022). The effect of transcranial direct current stimulation

(tDCS) combined with cognitive training on EEG spectral power in adolescent boys with ADHD: A double-blind, randomized, sham-controlled trial. *IBRO Neuroscience Reports*, 12, 55-64.

Kowalczyk, O.S., Mehta, M.A., O'Daly, O.G., & Criaud, M. (2021). Task-based functional connectivity in attention-deficit/hyperactivity disorder: A systematic review. *Biological Psychiatry Global Open Science*, 2(4), 350-367.

Kowalczyk, O.S., Pauls, A.M., Fusté, M., Williams, S.C.R., Hazelgrove, K., Vecchio, C., Seneviratne, G., Pariante, C.M., Dazzan, P., & Mehta, M.A. (2021). Neurocognitive correlates of working memory and emotional processing in postpartum psychosis: An fMRI study. *Psychological Medicine*, 51(10), 1724-1732.

Kowalczyk, O.S., Cubillo, A.I., Smith, A., Barrett, N., Giampietro, V., Brammer, M., Simmons, A., & Rubia, K. (2019). Methylphenidate and atomoxetine normalise fronto-parietal underactivation during sustained attention in ADHD adolescents. *European Neuropsychopharmacology*, 29(10), 1102-1116.

Wall, M.B., Pope, R., Freeman, T., **Kowalczyk, O.S.**, Demetriou, L., Mokrysz, C., Hindocha, C., Lawn, W., Bloomfield, M.A.P., Freeman, A.M., Feilding, A., Nutt, D.J., & Curran, H.V. (2019). Dissociable effects of cannabis with and without cannabidiol on the human brain's resting-state functional connectivity. *Journal of Psychopharmacology*, 33(7), 822-830.

Open scholarship

Kowalczyk, O.S., Lautarescu, A., Blok, E., Dall'Aglio, L., & Westwood, S.J. (2022). What senior academics can do to support reproducible and open research: A short, three-step guide. *BMC Research Notes*, 15(116).

SELECTED CONFERENCE ABSTRACTS

Characterising motor-related spinal cord activation: A test-retest reliability study.

2025 Organization for Human Brain Mapping, Brisbane, Australia

Assessment of non-protonated perflubron liquid-filled padding on cervical and lumbar spinal fMRI.

2025 Organization for Human Brain Mapping, Brisbane, Australia

A novel implementation of spinal fMRI demonstrates segmental organisation of functionally connected networks in the cervical spinal cord: A test-retest reliability study.

2022 International Association for the Study of Pain World Congress of Pain, Toronto, Canada

Neurocognitive effects of longer-term methylphenidate treatment on inhibitory function in boys with ADHD.

2022 Organization for Human Brain Mapping, Glasgow, UK

The effects of methylphenidate and atomoxetine on functional connectivity during working memory in boys with ADHD.

2021 British Association for Psychopharmacology, Online

Methylphenidate and atomoxetine normalise fronto-parietal activation in ADHD during sustained attention.

2018 EUNETHYDIS International Conference on ADHD, Edinburgh, UK

Neurocognitive correlates of working memory in postpartum psychosis.

2018 British Association for Psychopharmacology, London, UK

RESEARCH CULTURE

BNA Credibility Advisory Board Member British Neuroscience Association (BNA)	2023 – PRESENT
UKRN Local Network Lead for King's College London UK Reproducibility Network (UKRN)	2022 – PRESENT
Research and Innovation Committee Member IoPPN, King's College London	2022 – PRESENT
Personal Tutor IoPPN, King's College London	2021 – PRESENT
Postdoc Representative Department of Neuroimaging, IoPPN, King's College London	2021 – PRESENT
Steering Group Member The RIOT Science Club & King's Open Research Group Initiative	2018 – 2022
Diversity & Inclusion Committee Member Department of Neuroimaging, IoPPN, King's College London	2020 – 2021
PhD Student Representative Department of Child & Adolescent Psychiatry, IoPPN, King's College London	2018 – 2020